

Marine Microbial Biodiversity

Lecture 02

Microbial Physiology

Matteo Selci, PhD

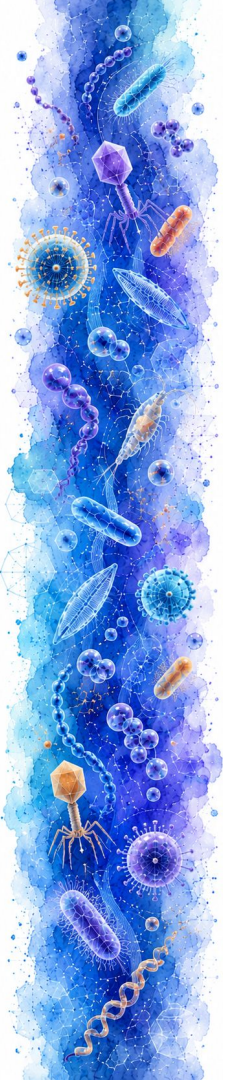
matteo.selci@unina.it

www.donatogiovannelli.com

@matteo.selci

@matteo_selci



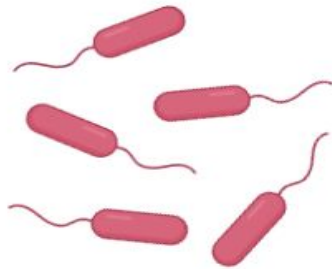


Link for the Marine Microbial Biodiversity slides

<https://github.com/MatS792/Marine-Microbiol-Biodiversity>

Microbial Cells

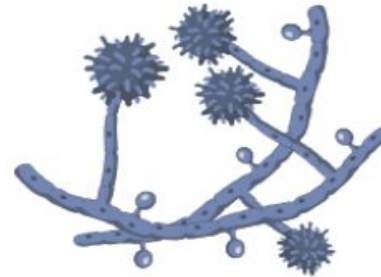
The classification of life into **three distinct domains** - **Bacteria**, **Archaea**, and **Eukarya** - has revolutionized our understanding of biological diversity. Bacteria and Archaea, both classified as **prokaryotes** due to their lack of membrane-bound nuclei, represent **two domains** that, despite structural similarities, are **genetically and biochemically distinct**. Archaea often thrive in extreme environments, exhibiting unique metabolic pathways and membrane compositions that distinguish them from Bacteria. **Eukarya**, comprising **all organisms with complex, membrane-bound organelles**, includes familiar life forms such as **animals, plants, fungi, and protists**. This tripartite system underscores the **profound evolutionary divergences** that have shaped cellular life, providing a framework for studying **life's origin** and the **evolutionary relationships among all organisms**.



Bacteria

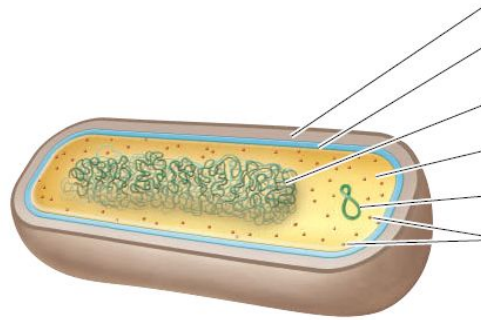


Archaea

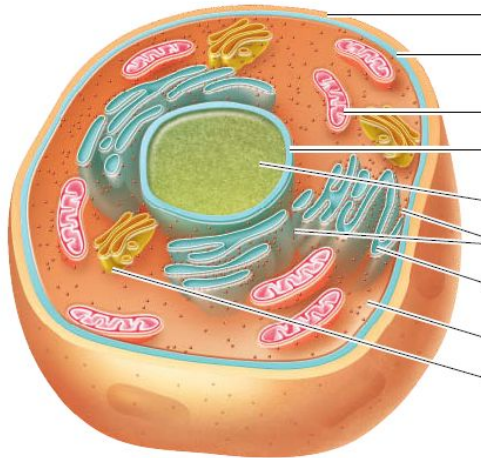


Eukarya

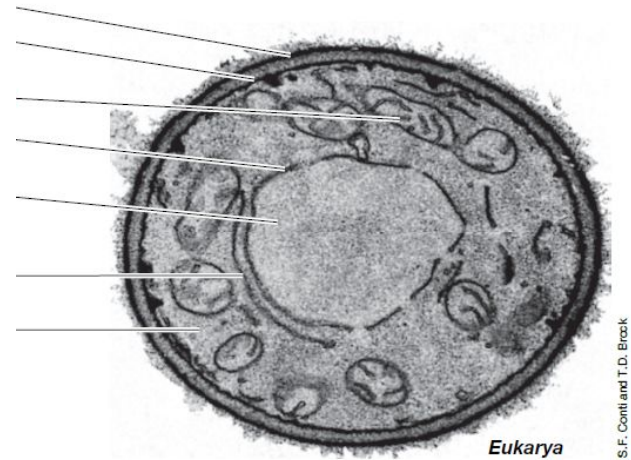
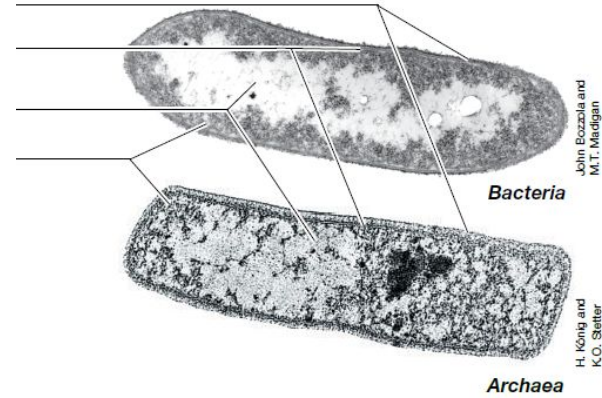
Cellular Structures



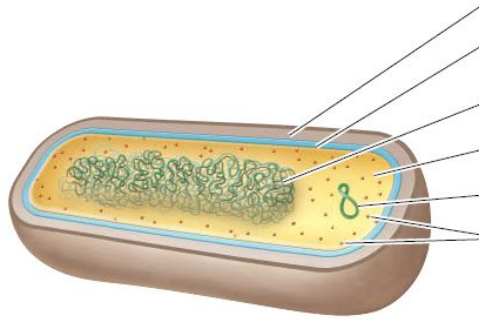
(a) Prokaryotic cell



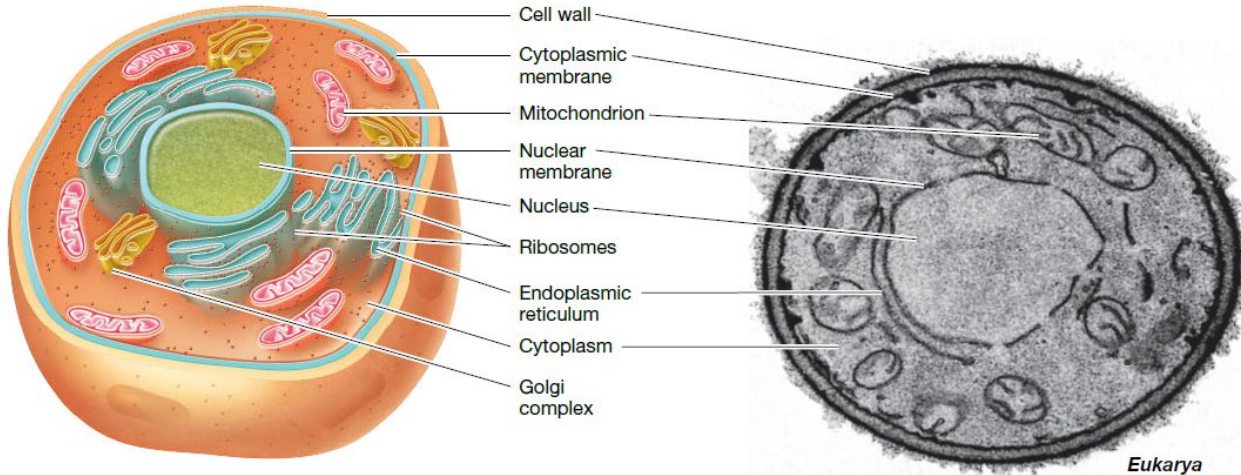
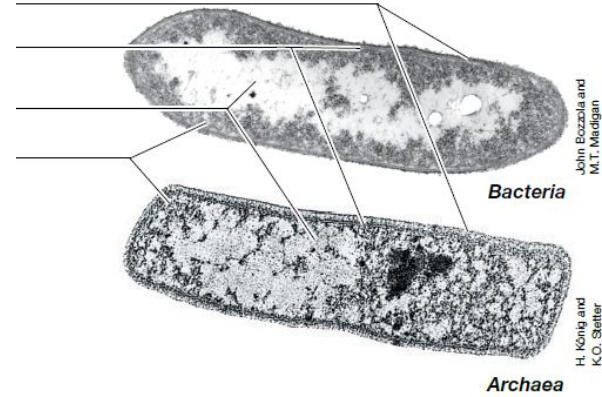
(b) Eukaryotic cell



Cellular Structures

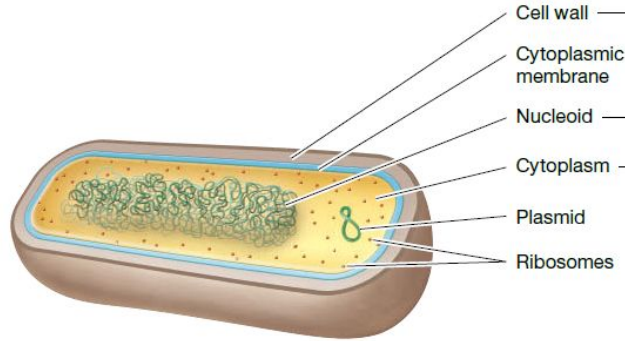


(a) Prokaryotic cell

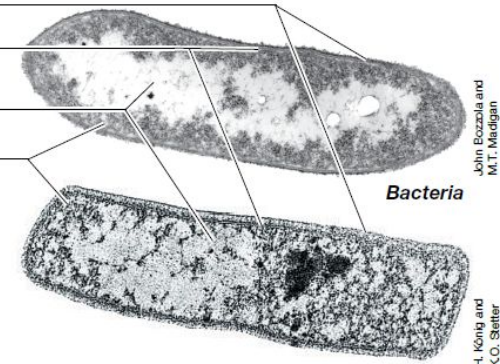


(b) Eukaryotic cell

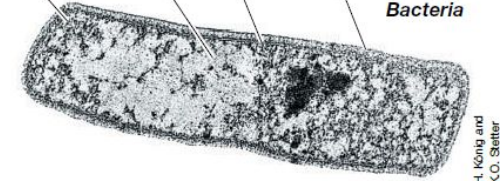
Cellular Structures



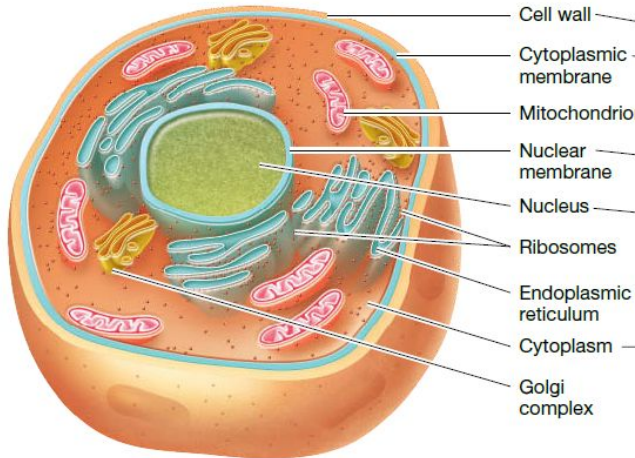
(a) Prokaryotic cell



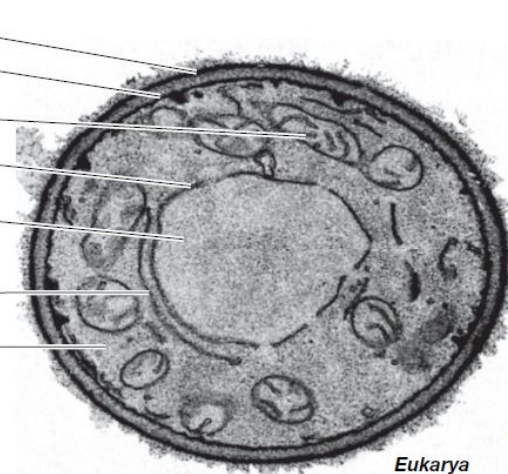
Bacteria



Archaea



(b) Eukaryotic cell



Eukarya

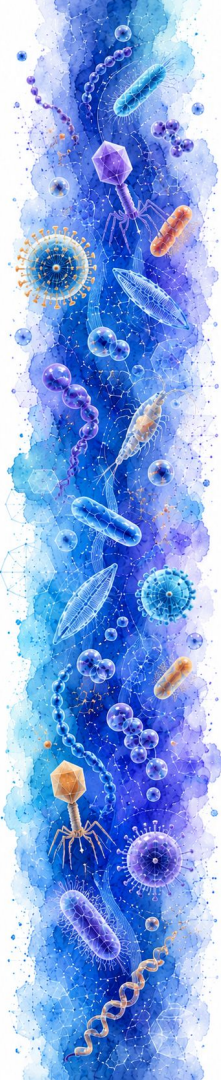


Table 12.1 Major structural and physiological characteristics of Bacteria, Archaea, and Eukarya^a

<i>Characteristic</i>	Bacteria	Archaea	Eukarya
Morphological			
Prokaryotic cell structure	Yes	Yes	No
Cell wall	Peptidoglycan present	No peptidoglycan	No peptidoglycan
Membrane lipids	Ester-linked	Ether-linked	Ester-linked
Membrane-enclosed nucleus	Absent	Absent	Present
Flagella mechanism	Rotation	Rotation	Whiplike
Sensitivity to chloramphenicol, streptomycin, kanamycin, and penicillin	Yes	No	No
Physiological/special structures			
Dissimilative reduction of S ⁰ or SO ₄ ²⁻ to H ₂ S, or Fe ³⁺ to Fe ²⁺	Yes	Yes	No
Nitrification (ammonia oxidation)	Yes	Yes	No
Chlorophyll-based photosynthesis	Yes	No	Yes (in chloroplasts)
Denitrification	Yes	Yes	No
Nitrogen fixation	Yes	Yes	No
Rhodopsin-based energy metabolism	Yes	Yes	No
Chemolithotrophy (Fe ²⁺ , NH ₃ , S ⁰ , H ₂)	Yes	Yes	No
Endospores	Yes	No	No
Gas vesicles	Yes	Yes	No
Storage granules of poly-β-hydroxyalkanoates	Yes	Yes	No
Growth above 70°C	Yes	Yes	No
Growth above 100°C	No	Yes	No

^aNote that for many features, only particular representatives within a domain show the property.

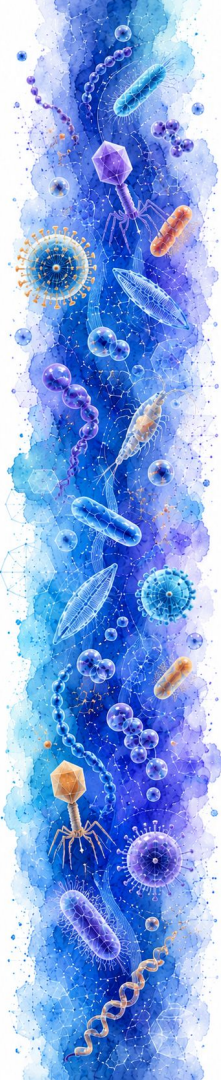


Table 12.1 Major structural and physiological characteristics of Bacteria, Archaea, and Eukarya^a

<i>Characteristic</i>	Bacteria	Archaea	Eukarya
Morphological			
Prokaryotic cell structure	Yes	Yes	No
Cell wall	Peptidoglycan present	No peptidoglycan	No peptidoglycan
Membrane lipids	Ester-linked	Ether-linked	Ester-linked
Membrane-enclosed nucleus	Absent	Absent	Present
Flagella mechanism	Rotation	Rotation	Whiplike
Sensitivity to chloramphenicol, streptomycin, kanamycin, and penicillin	Yes	No	No
Physiological/special structures			
Dissimilative reduction of S ⁰ or SO ₄ ²⁻ to H ₂ S, or Fe ³⁺ to Fe ²⁺	Yes	Yes	No
Nitrification (ammonia oxidation)	Yes	Yes	No
Chlorophyll-based photosynthesis	Yes	No	Yes (in chloroplasts)
Denitrification	Yes	Yes	No
Nitrogen fixation	Yes	Yes	No
Rhodopsin-based energy metabolism	Yes	Yes	No
Chemolithotrophy (Fe ²⁺ , NH ₃ , S ⁰ , H ₂)	Yes	Yes	No
Endospores	Yes	No	No
Gas vesicles	Yes	Yes	No
Storage granules of poly-β-hydroxyalkanoates	Yes	Yes	No
Growth above 70°C	Yes	Yes	No
Growth above 100°C	No	Yes	No

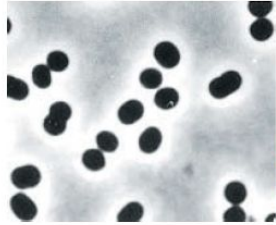
^aNote that for many features, only particular representatives within a domain show the property.

Prokaryotic Morphology and Size

coccus (1.5 μm)



Coccus
(cocci)



Norbert Pfennig



Rod
(rods)



Norbert Pfennig

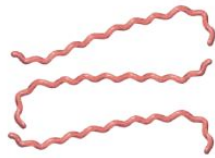
spirillum (1 μm)



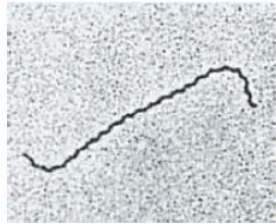
Spirillum
(spirilla)



Norbert Pfennig



Spirochete



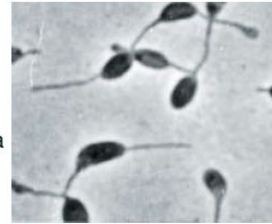
E. Canale-Parola



Stalk

Hypha

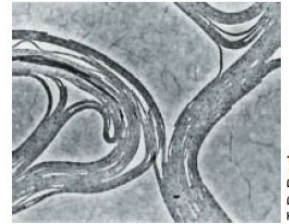
Budding and
appendaged



Norbert Pfennig



Filamentous



T. D. Brock

spirochete (0.25 μm)

budding (1.2 μm)

filamentous (0.8 μm)

Reported dimensions represent **cell diameters** in phase-contrast photomicrographs

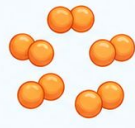
Prokaryotic Morphology and Size

Bacterial Shapes and Arrangements

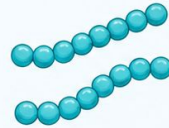
Spherical forms and arrangements



Cocci
spherical cells



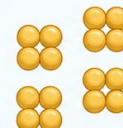
Diplococci
pairs



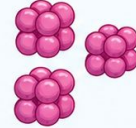
Streptococci
chains



Staphylococci
grape-like clusters



Tetrads
groups of four



Sarcina
cubical packets

Rod-shaped and curved forms



Bacilli
rod-shaped cells



Coccobacilli
short rods



Vibrios
comma-shaped cells

Spiral, filamentous and variable forms



Spirilla
rigid spirals



Spirochetes
flexible corkscrews



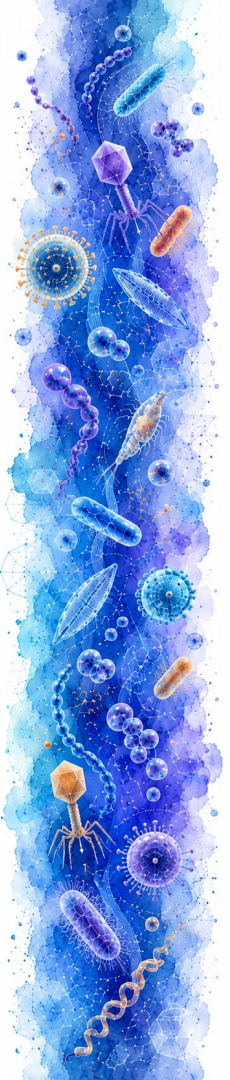
Filamentous bacteria
elongated filaments



Pleomorphic bacteria
variable shapes

Simplified morphological overview — shapes and arrangements may vary among species

Reported dimensions represent **cell diameters** in phase-contrast photomicrographs



Prokaryotic Morphology and Size

Although cell morphology is easily determined, it is a poor predictor of other properties of a cell.

For example: under the microscope many rod-shaped *Archaea* look identical to rod-shaped *Bacteria*, yet we know they are of different phylogenetic domains.

Prokaryotic Morphology and Size

Although cell morphology is easily determined, it is a poor predictor of other properties of a cell.
For example: under the microscope many rod-shaped *Archaea* look identical to rod-shaped *Bacteria*, yet we know they are of different phylogenetic domains.



Dimetrodon (294
to 271 mya)



Spinosaurus
(97 to 95 mya)

170 to 195 million years

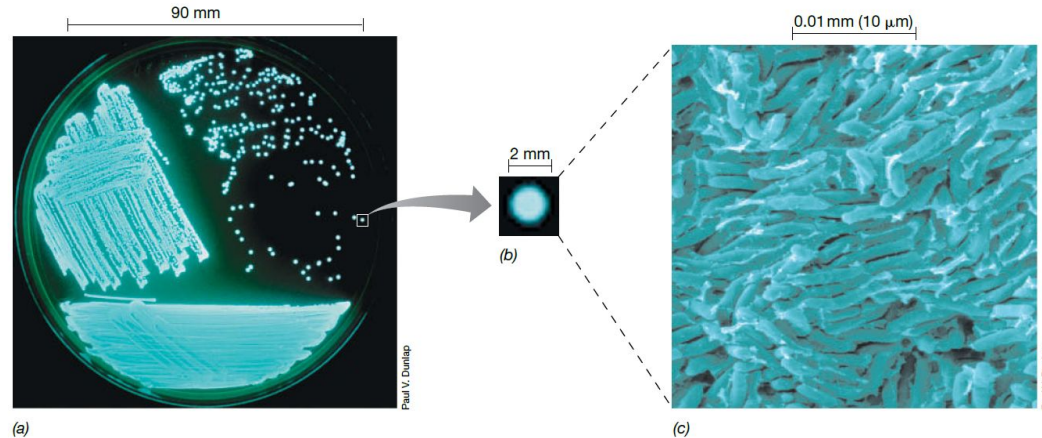
Prokaryotic Morphology and Size

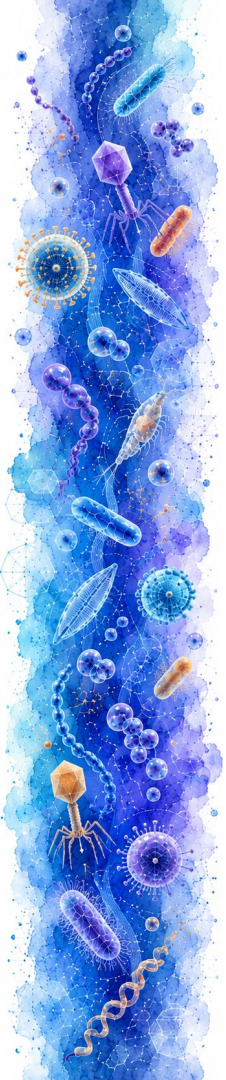
Although cell morphology is easily determined, it is a poor predictor of other properties of a cell.

For example: under the microscope many rod-shaped *Archaea* look identical to rod-shaped *Bacteria*, yet we know they are of different phylogenetic domains.

Prokaryotes vary in size from cells smaller of about $0.2\ \mu\text{m}$ in diameter to those more than $700\ \mu\text{m}$.

The vast majority of rod-shaped prokaryotes that have been cultured are between 0.5 and $4\ \mu\text{m}$ wide and less than $15\ \mu\text{m}$ long.





Prokaryotic Morphology and Size

Although cell morphology is easily determined, it is a poor predictor of other properties of a cell.

For example: under the microscope many rod-shaped *Archaea* look identical to rod-shaped *Bacteria*, yet we know they are of different phylogenetic domains.

Prokaryotes vary in size from cells smaller of about **0.2 μm** in diameter to those more than **700 μm** .

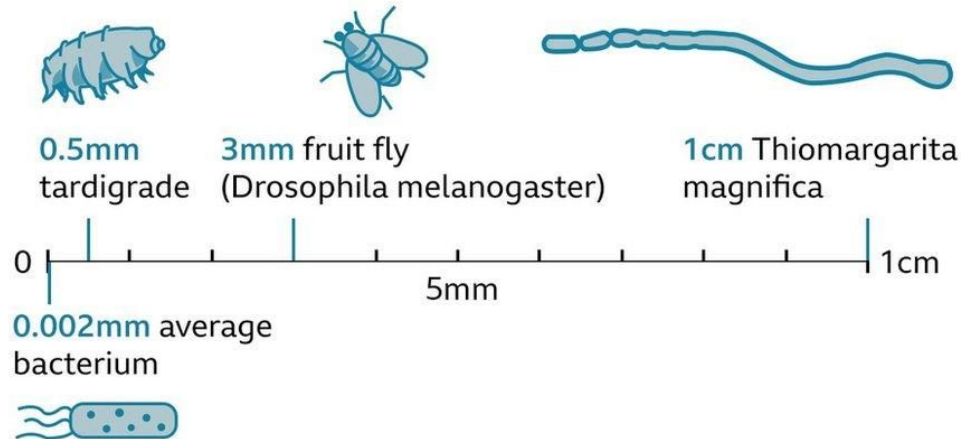
The vast majority of rod-shaped prokaryotes that have been cultured are between 0.5 and 4 μm wide and less than 15 μm long.

YES BUT...

Prokaryotic Morphology and Size

The **GIANT** *Thiomargarita magnifica* (Volland et al., 2022)

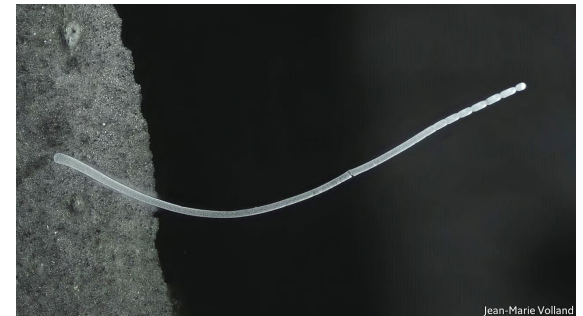
How does the big bacterium compare?



Source: Science

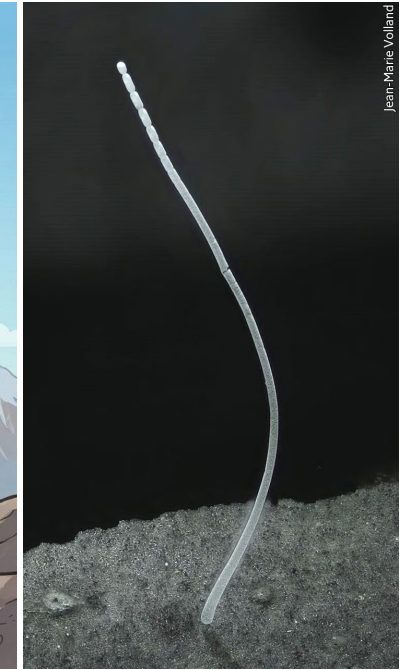
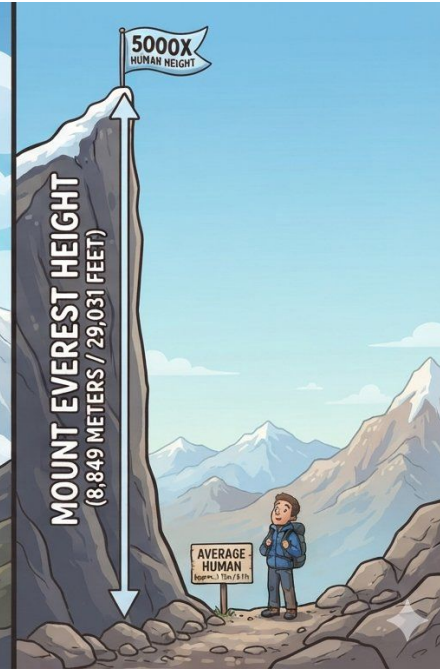
B B C

5000 times bigger than the smallest bacteria



Prokaryotic Morphology and Size

The **GIANT** *Thiomargarita magnifica* (Volland et al., 2022)



5000 time bigger than the smallest bacteria

Prokaryotic Morphology and Size

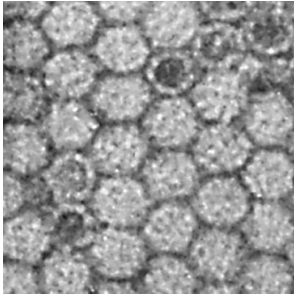
Table 1.1 Size range of some representative marine microbes

Organism	Characteristics	Size (μm) ^a	Volume (μm^3) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliana huxleyi</i>	0.17	0.003
<i>Thermodiscus</i>	Disc-shaped. Hyperthermophilic archaeon	0.2×0.08	0.003
" <i>Ca. Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1×0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1×2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5×80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca auraucae</i>	Filamentous. Sulfur bacterium	30×43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50×160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80×600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	141 371 00

^aApproximate diameter $\times$ length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.

Prokaryotic Morphology and Size

Table 1.1 Size range of some representative marine microbes



Brevidensovirus is a member of the *Parvoviridae* family, a group of small, icosahedral viruses with single-stranded DNA genomes. These viruses typically infect marine invertebrates, such as crustaceans, causing disease outbreaks. This can influence food webs and ecosystem equilibria.

Organism	Characteristics	Size (μm) ^a	Volume (μm ³) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliania huxleyi</i>	0.17	0.003
<i>Thermoplasma</i>	Disc-shaped. Hyperthermophilic archaeon	0.2 × 0.08	0.003
" <i>Ca. Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1 × 0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1 × 2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5 × 80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca auraucae</i>	Filamentous. Sulfur bacterium	30 × 43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50 × 160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80 × 600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	141 371 00

^aApproximate diameter × length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.

Prokaryotic Morphology and Size

Table 1.1 Size range of some representative marine microbes



Pelagibacter ubique is one of the most abundant and ecologically significant bacteria in marine environments. It belongs to the SAR11 clade, a group of Alphaproteobacteria that dominates the surface waters of oceans worldwide.

Organism	Characteristics	Size (μm) ^a	Volume (μm^3) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliania huxleyi</i>	0.17	0.003
<i>Thermodiscus</i>	Disc-shaped. Hyperthermophilic archaeon	0.2×0.08	0.003
"Ca. <i>Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1×0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1×2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5×80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca auraucae</i>	Filamentous. Sulfur bacterium	30×43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50×160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80×600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	141 371 00

^aApproximate diameter $\times$ length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.

Prokaryotic Morphology and Size

Ostreococcus is a genus of unicellular green algae and is considered one of the smallest known free-living eukaryotes.



It plays a crucial role in primary production, particularly in oligotrophic waters. As a photosynthetic organism, it contributes to the fixation of carbon dioxide forming the base of the marine food web.

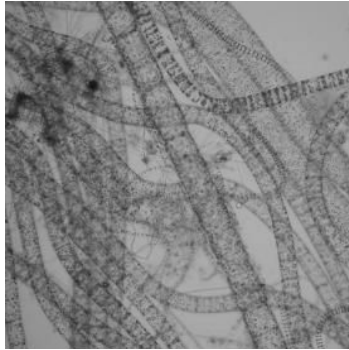
Table 1.1 Size range of some representative marine microbes

Organism	Characteristics	Size (μm) ^a	Volume (μm^3) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliania huxleyi</i>	0.17	0.003
<i>Thermodiscus</i>	Disc-shaped. Hyperthermophilic archaeon	0.2×0.08	0.003
" <i>Ca. Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1×0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1×2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5×80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca auraucae</i>	Filamentous. Sulfur bacterium	30×43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50×160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80×600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	14137100

^aApproximate diameter $\times$ length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.

Prokaryotic Morphology and Size

Table 1.1 Size range of some representative marine microbes



Thioploca araucae is a sulfur-oxidizing bacterium found in marine environments, particularly in oxygen-depleted, sulfide-rich sediments along continental shelves. It contributes to the detoxification of sulfide-rich environments and plays a vital role in the sulfur and nitrogen cycles in marine ecosystems.

Organism	Characteristics	Size (μm) ^a	Volume (μm ³) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliania huxleyi</i>	0.17	0.003
<i>Thermoplasma</i>	Disc-shaped. Hyperthermophilic archaeon	0.2 × 0.08	0.003
" <i>Ca. Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1 × 0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1 × 2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5 × 80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca araucae</i>	Filamentous. Sulfur bacterium	30 × 43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50 × 160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80 × 600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	141 371 00

^aApproximate diameter × length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.

Prokaryotic Morphology and Size

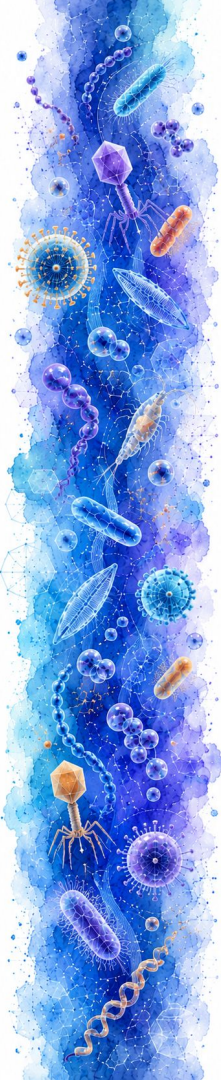
Table 1.1 Size range of some representative marine microbes

Organism	Characteristics	Size (μm) ^a	Volume (μm ³) ^b
<i>Brevidensovirus</i>	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliana huxleyi</i>	0.17	0.003
<i>Thermodiscus</i>	Disc-shaped. Hyperthermophilic archaeon	0.2 × 0.08	0.003
" <i>Ca. Pelagibacter ubique</i> " ^c	Crescent-shaped bacterium ubiquitous in ocean plankton (cultured example of SAR11 clade)	0.1 × 0.9	0.01
<i>Megavirus chilensis</i>	Giant virus infecting marine amoebae	0.44	0.045
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga. Smallest known eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Bacteria common in coastal environments and associated with animals and human diseases	1 × 2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom which produces toxic domoic acid	5 × 80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Thioploca auraucae</i>	Filamentous. Sulfur bacterium	30 × 43	30000
<i>Lingulodinium polyedra</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Beggiatoa</i>	Filamentous. Sulfur bacterium	50 × 160	314000
<i>Epulopiscium fishelsoni</i>	Rods. Bacteria symbiotic in fish gut	80 × 600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300 ^d	141 371 00



Thiomargarita namibiensis is a species of sulfur bacteria, known for being one of the largest bacteria ever discovered. It can reach sizes of up to 0.75 mm in diameter, making it visible to the naked eye. Found in ocean sediments off the coast of Namibia, belongs to a group of bacteria called *Beggiatoaceae*, which are known for their role in sulfur and nitrogen cycling.

^aApproximate diameter × length; where one value is given, this is the diameter of spherical virus particles or cells. ^bApproximate values calculated assuming spherical or cylindrical shapes. ^c*Candidatus*, provisional taxonomic name—see Chapter 4. ^dCells up to 750 μm have been recorded.



But Size Really Matters?

But Size Really Matters?

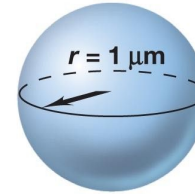
The lower limit is controlled by volume needed to house the essential components of a free-living cell, calculated around $0.1 \mu\text{m}$ in diameter.

The upper limit is controlled by diffusion limitation of nutrients and waste from and to the surrounding environment.

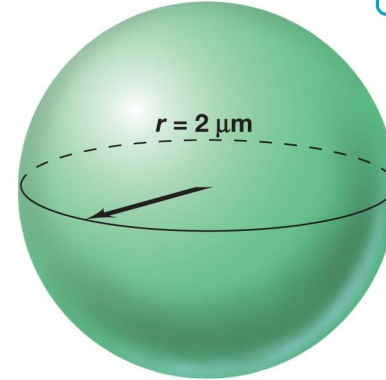
At the microscale, **the rate of molecular diffusion becomes the most important mechanism for transport of substances into and out of the cell.** Small cells feeding by absorption (osmotrophy) can take up nutrients more efficiently than larger cells.

The surface area to volume ratio (SA/V) is the critical factor because as cell size increases, volume (V) increases more quickly than surface area (SA).

Higher the SA/V, larger is the surface area per unit volume through which material can diffuse, faster is the diffusion process.



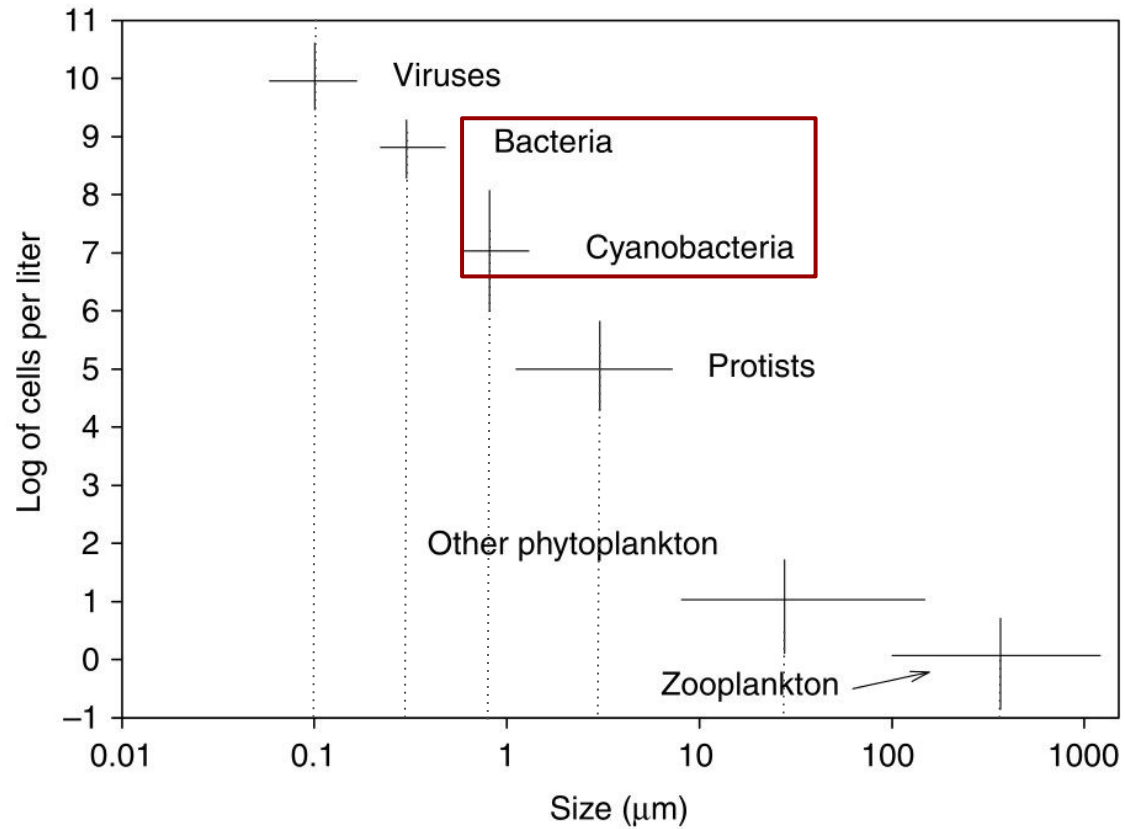
$$\begin{aligned} r &= 1 \mu\text{m} \\ \text{Surface area } (4\pi r^2) &= 12.6 \mu\text{m}^2 \\ \text{Volume } (\frac{4}{3}\pi r^3) &= 4.2 \mu\text{m}^3 \\ \frac{\text{Surface}}{\text{Volume}} &= 3 \end{aligned}$$



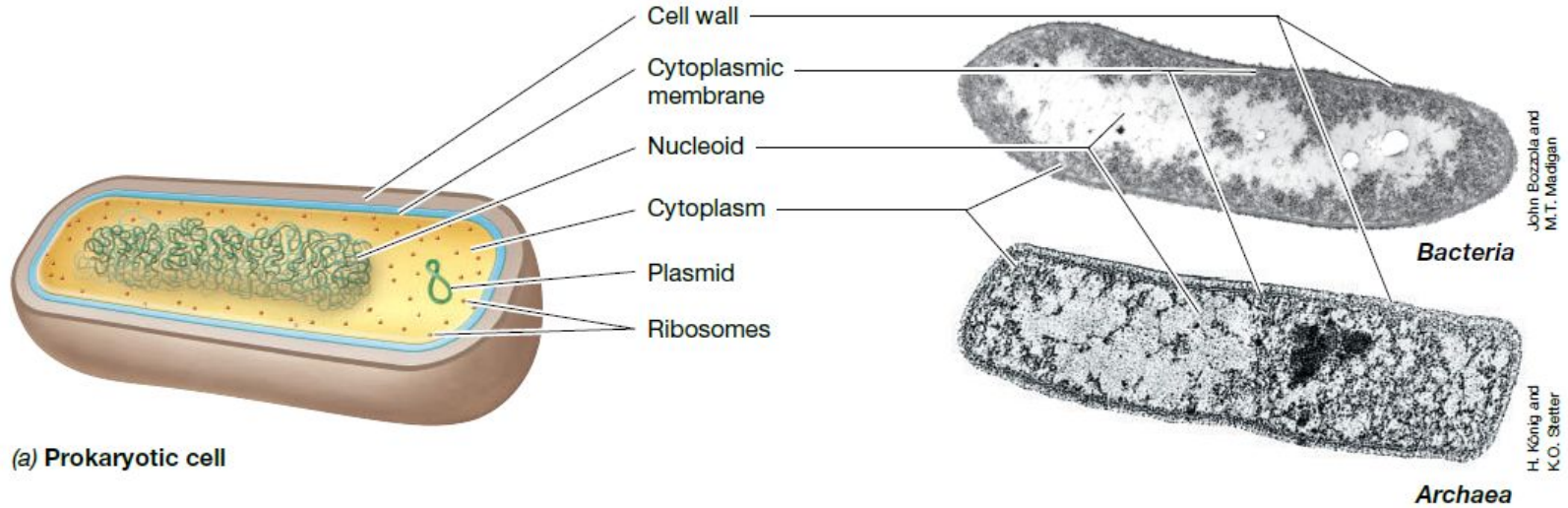
$$\begin{aligned} r &= 2 \mu\text{m} \\ \text{Surface area} &= 50.3 \mu\text{m}^2 \\ \text{Volume} &= 33.5 \mu\text{m}^3 \\ \frac{\text{Surface}}{\text{Volume}} &= 1.5 \end{aligned}$$

Small size in marine environment is a strategy to maximize the uptake of nutrients across the cell membrane

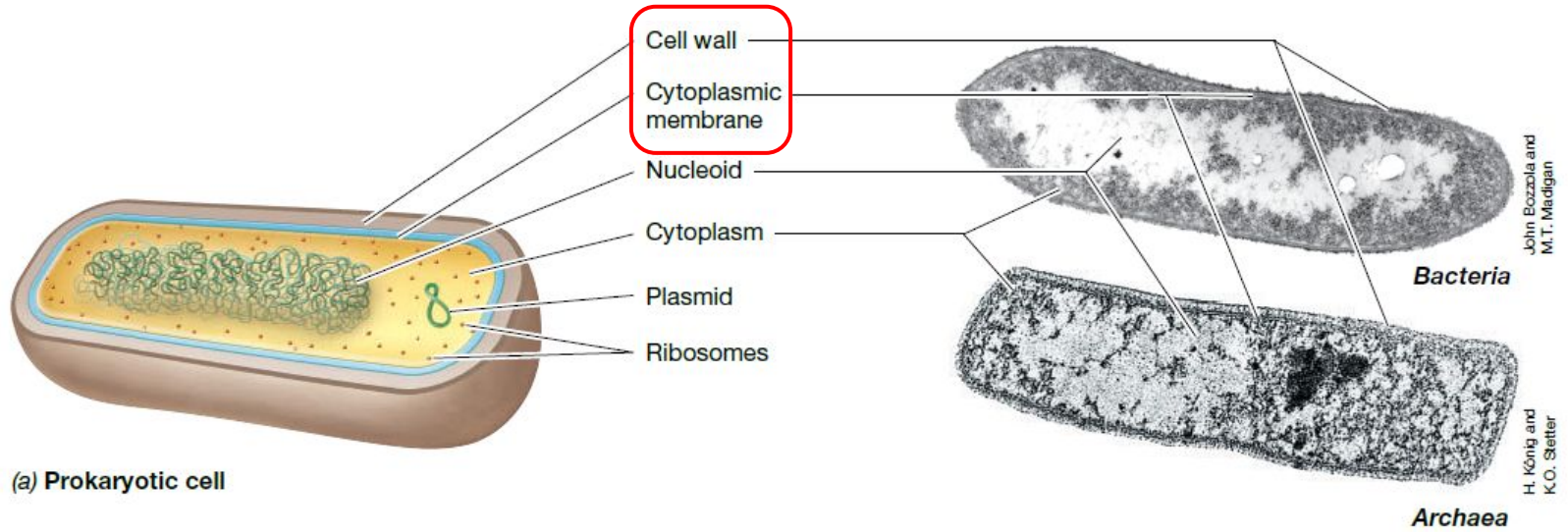
Abundance And Size



Prokaryotic Microbial Cell

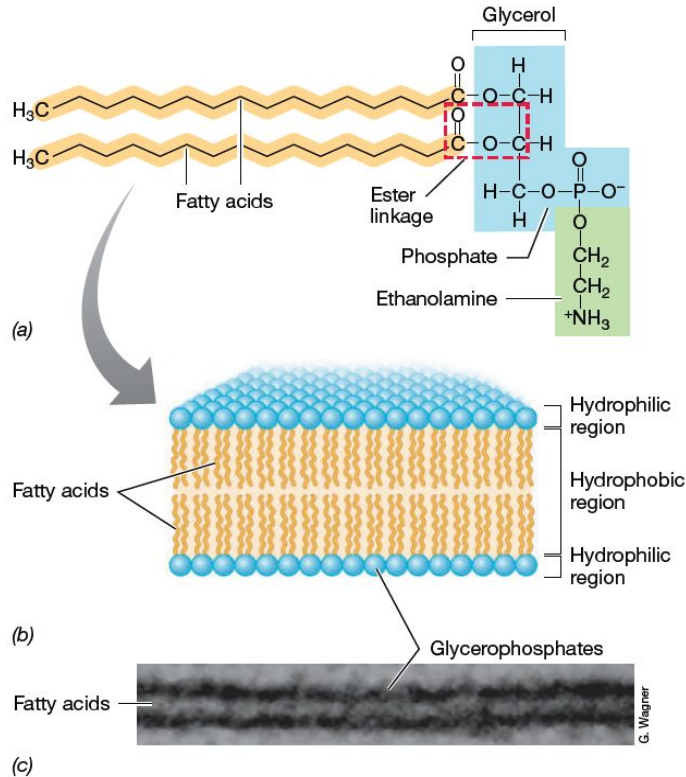


Prokaryotic Microbial Cell

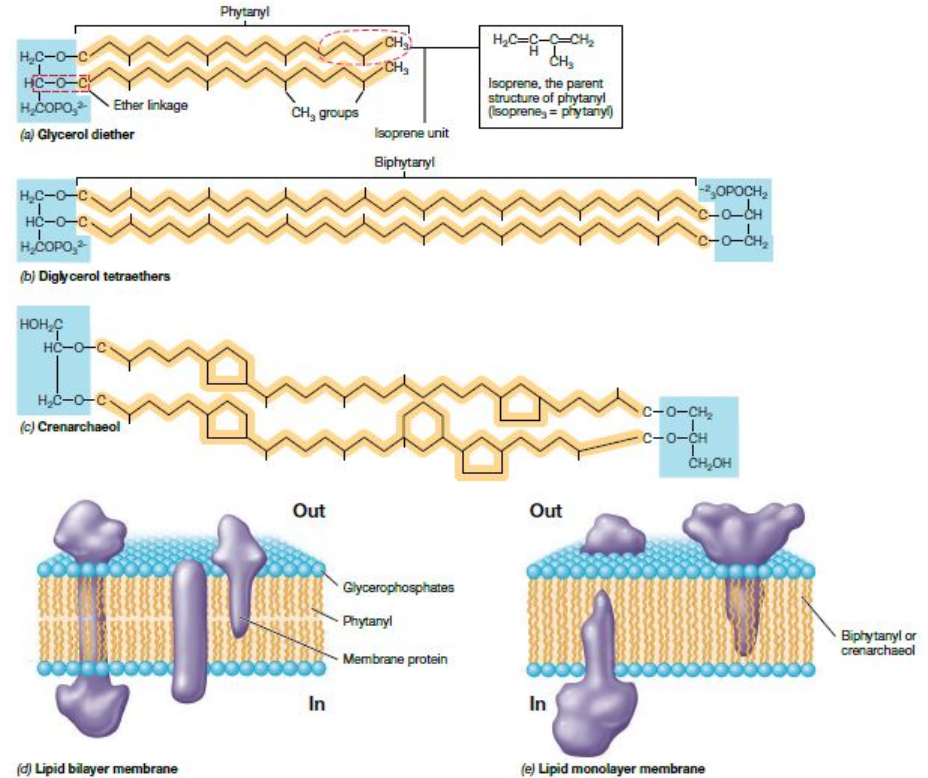


Prokaryotic Microbial Cell

Bacteria

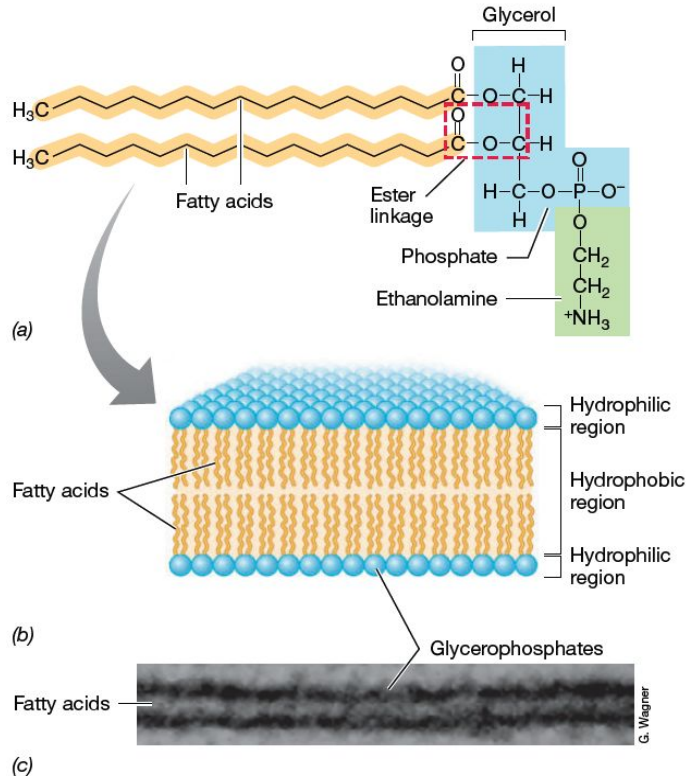


Archaea



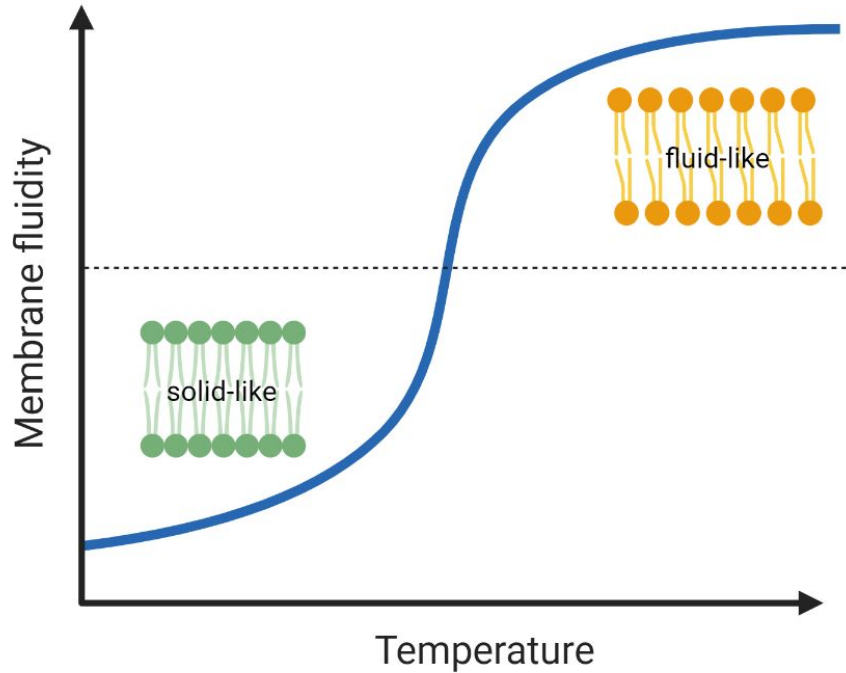
Prokaryotic Microbial Cell

Bacteria



The **bacterial cell membrane** consists of a phospholipid bilayer acting as a semi-permeable selective barrier, controlling the entry and exit of substances. Each phospholipid molecule is composed by a hydrophilic "head" and two hydrophobic "tails". In detail, two fatty acid tails are connected to one glycerol unit of the backbone via **ester bonds**. This occurs between the hydroxyl groups of glycerol and the carboxyl groups of fatty acids. The ester linkages contribute to the flexibility of the membrane. It can host various proteins involved in transport, energy production, and signal transduction. The bacterial cell wall is mainly made of **peptidoglycan**, a polymer composed of sugars and amino acids. This structure provides rigidity and shape to the cell. In **Gram-positive** bacteria, the cell wall is **thick** and consists of **multiple layers of peptidoglycan**, along with **teichoic acids**. **Gram-negative** bacteria have a **thinner peptidoglycan** layer sandwiched **between an inner cytoplasmic membrane and an outer membrane** that contains lipopolysaccharides (LPS).

Prokaryotic Microbial Cell



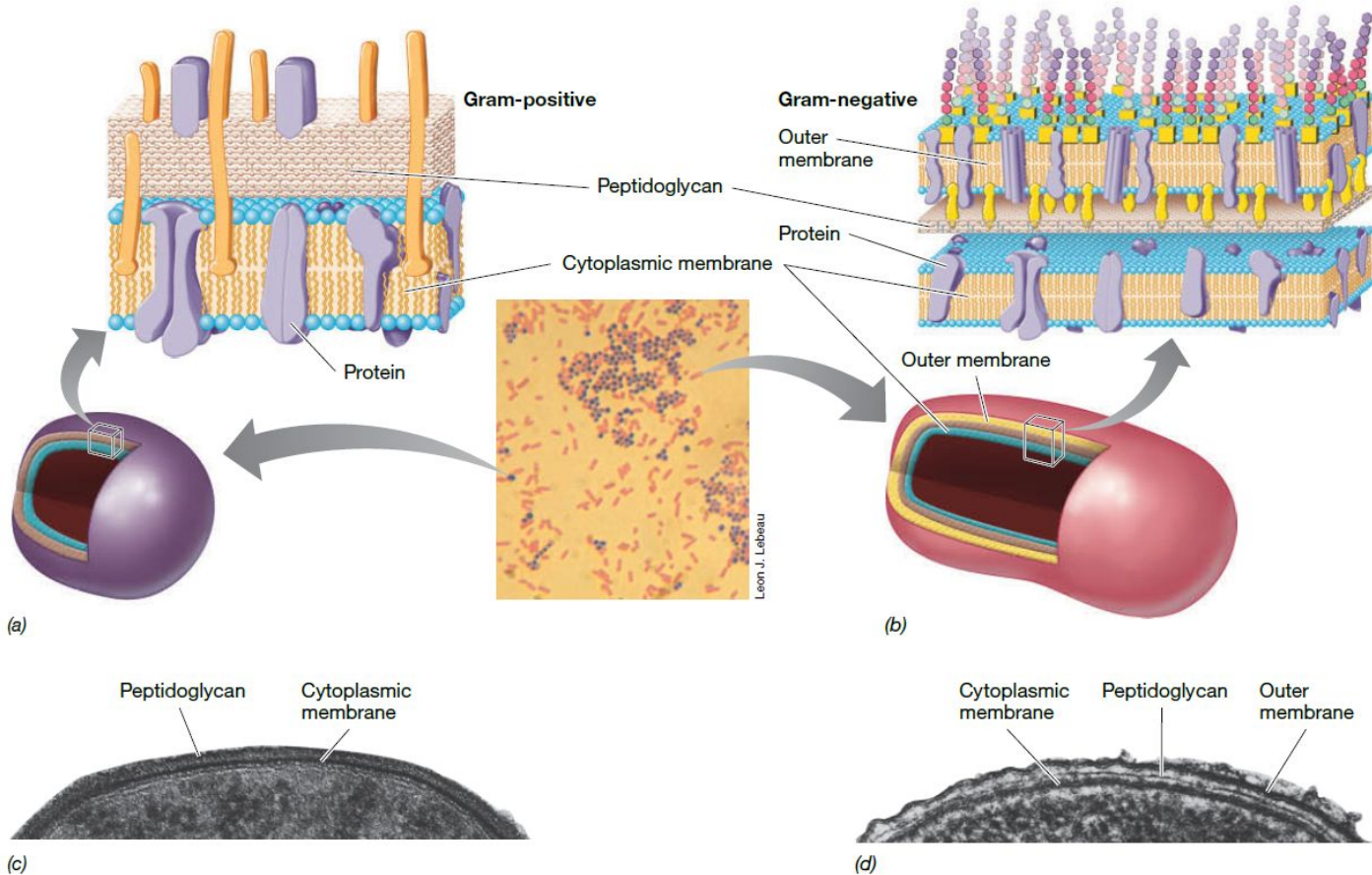
fluid-like

flexible state of membrane
unpacked and disorganized phospholipids
high permeability

solid-like

rigid state of membrane
closely packed phospholipids
low permeability

Bacterial Cell walls

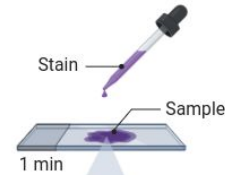


Gram Staining Protocol

Step 1

Crystal violet

Primary stain added to specimen smear.

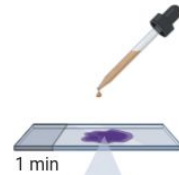


● Gram (+): purple
● Gram (-): purple

Step 2

Iodine

Mordant makes dye less soluble so it adheres to cell walls.

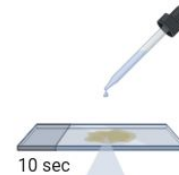


● Gram (+): purple
● Gram (-): purple

Step 3

Alcohol

Decolorizer washes away stain from gram (-) cell walls.

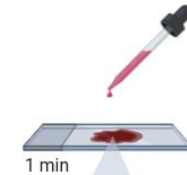


● Gram (+): purple
● Gram (-): colorless

Step 4

Safranin

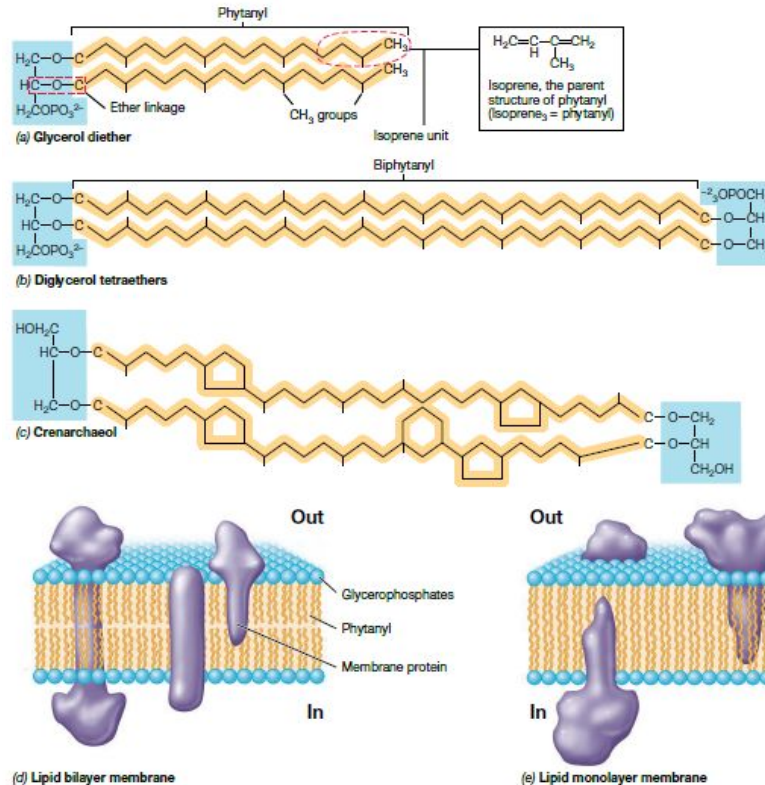
Counterstain allows dye adherence to gram (-) cell walls.



● Gram (+): purple
● Gram (-): red

Prokaryotic Microbial Cell

Archaea



Archaeal cell membranes are primarily composed of phospholipids in which the fatty acid chains (isoprenoids) are connected to glycerol through **ether bonds**, which are **more stable than ester bonds** found in bacterial and eukaryotic membranes. Some archaeal species, particularly those in extreme environments (like hyper-thermophiles), possess a **monolayer membrane** structure, where the hydrophobic tails of the lipids are linked together, forming a single layer. Archaeal phospholipids typically contain branched isoprenoid chains, which provide greater resistance to heat. Common lipids include **phytanyl** and **biphytanyl** (or crenarchaeol).

Archaeal cell walls lack **peptidoglycan**. Instead, they may contain **pseudomurein** (similar to peptidoglycan but with different sugar components) or other **polysaccharides** and **proteins**. Some archaea have a cell wall composed of a **protein** or **glycoprotein** layer known as a **S-layer**.

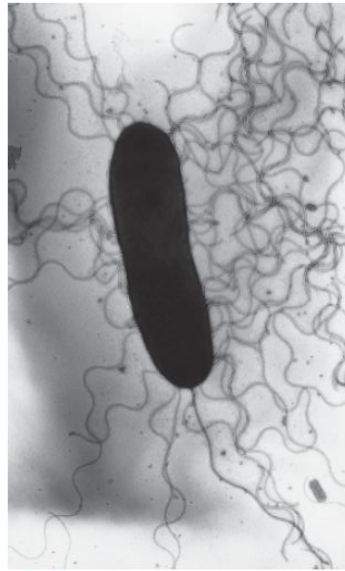
Prokaryotic Cell Structure And Inclusions



Frank Dazzo and Richard Heinzen

Protection

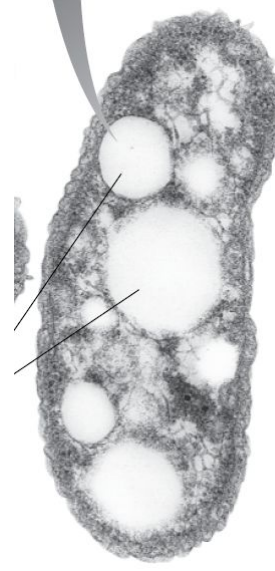
Capsule



Carl E. Bauer

Motility

Flagella



F. R. Turner and M. T. Madigan

Storage

Polyhydroxyalkanoates (PHAs)



H. Hippe

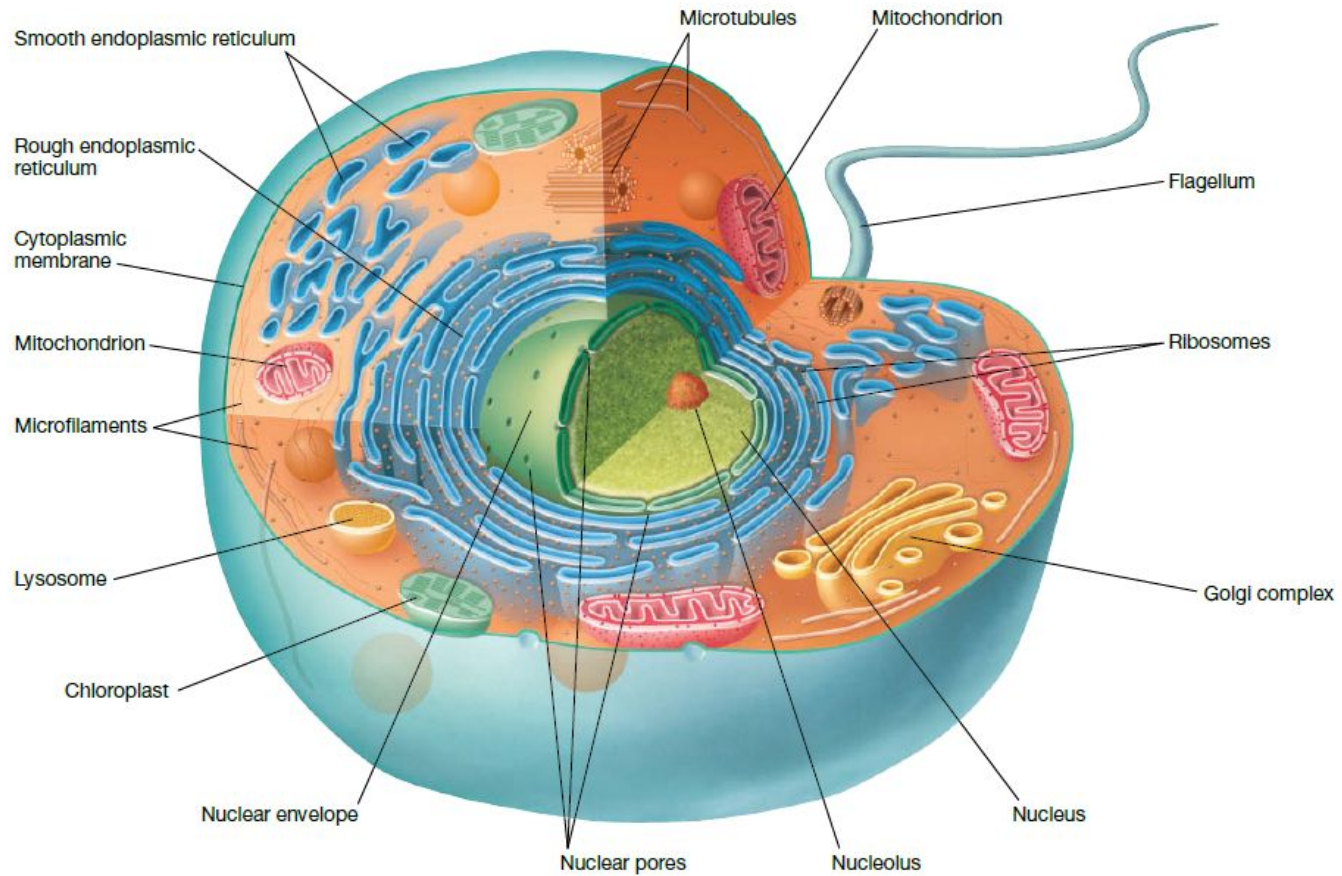
Survival

Endospore

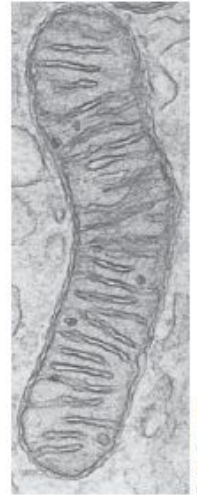
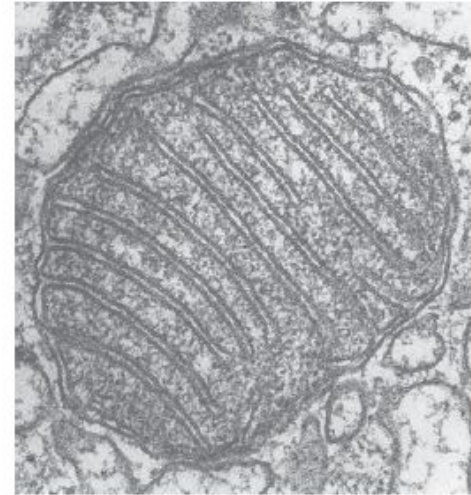
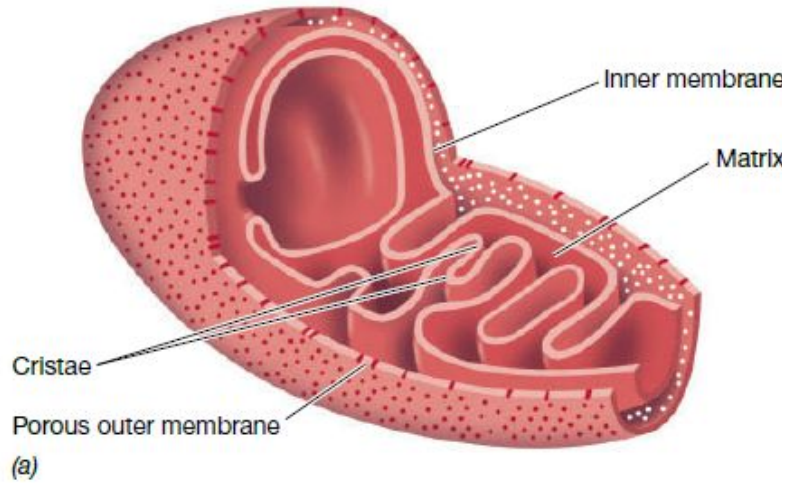


H. Hippe

Eukaryotic Cell Structure



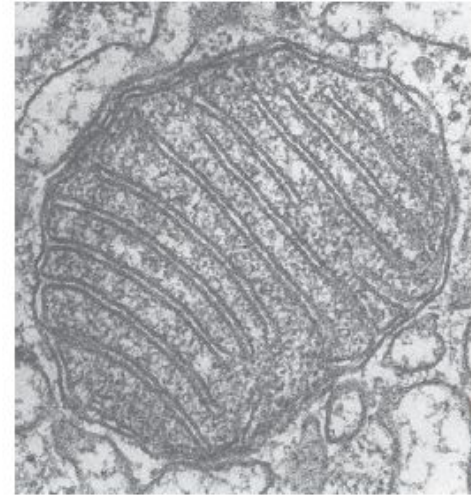
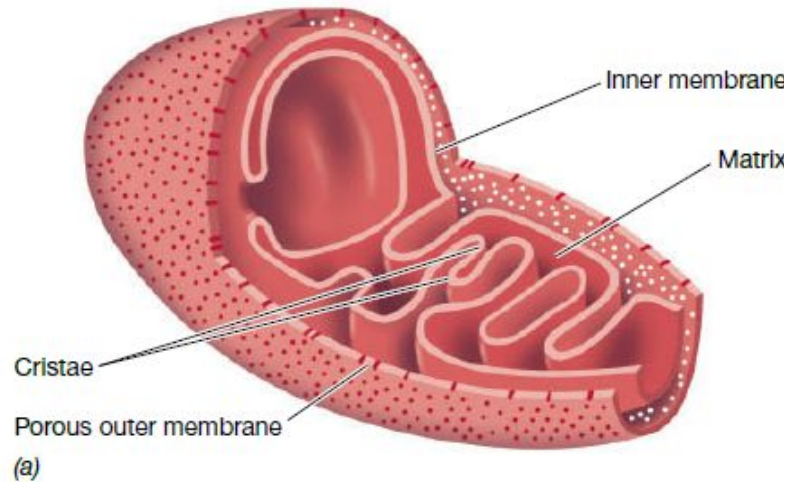
Eukaryotic Organelles



D. W. Fawcett

D. W. Fawcett

Eukaryotic Organelles

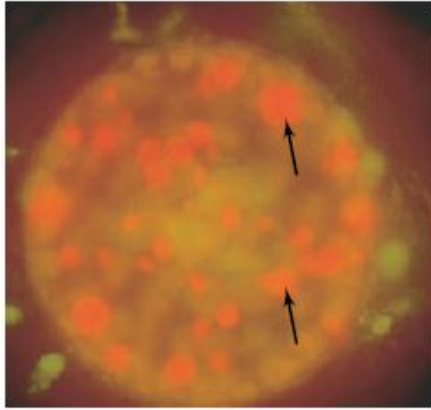


Mitochondria is where respiration occurs in aerobic eukaryotic cells.

Cristae contains the enzymes involved in respiration and **ATP production**.

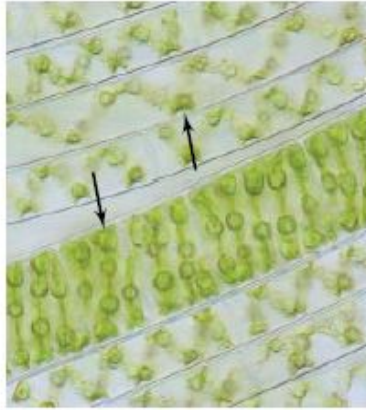
Matrix contains enzymes of the citric acid cycle, for the oxidation of organic compounds to CO_2 .

Eukaryotic Organelles

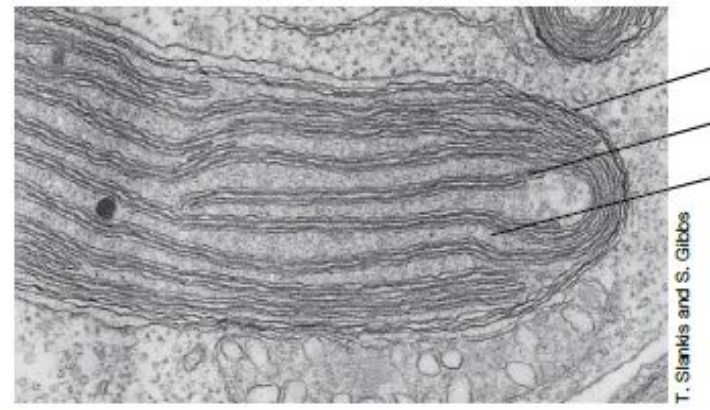


T. D. Brock

(a)



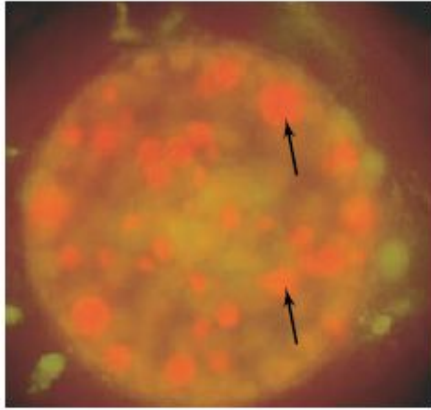
(b)



T. Slankis and S. Gibbs

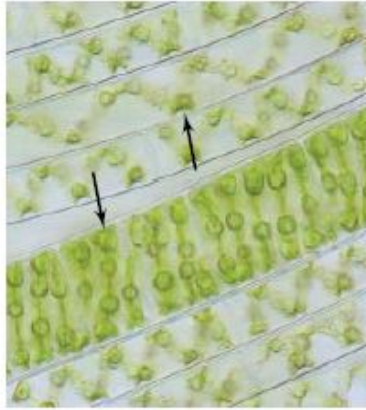
(c)

Eukaryotic Organelles

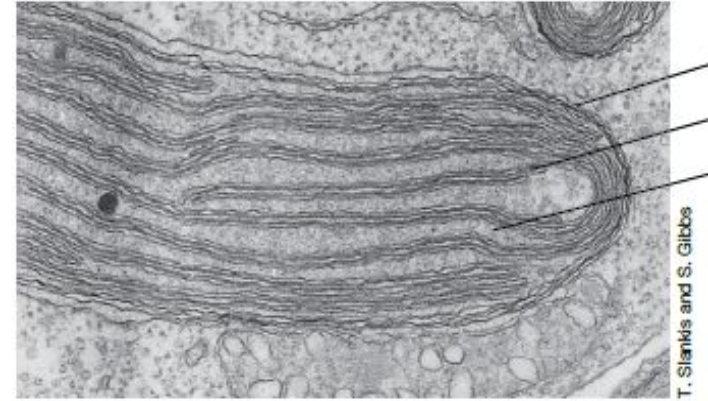


T. D. Brock

(a)



(b)



T. Slankis and S. Gibbs

(c)

Chloroplasts is where photosynthesis occurs in phototrophic microbial eukaryotes. Thylakoids contain chlorophyll and all other components needed for ATP synthesis. Stroma contains the ribulose biphosphate carboxylase (RubisCO), key enzyme of the Calvin cycle for CO_2 conversion to organic compounds.

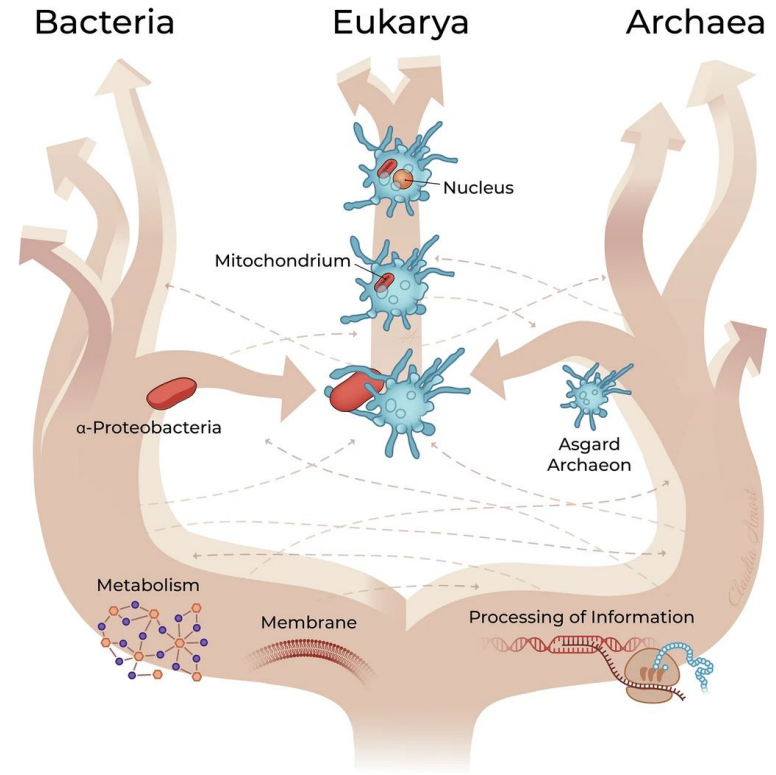
Endosymbiosis And Eukaryotic Cell Evolution

The Serial Endosymbiotic Theory, proposed by American biologist Lynn Margulis in the late 1960s, explains that complex eukaryotic cells evolved through the **sequential merging** of independent prokaryotic bacteria

1° Merger: An ancestral archaeon-like host cell merged with or engulfed an energy-producing, aerobic proteobacterium, which eventually evolved into the mitochondrion

2° Merger: In lineages that led to plants and algae, a subsequent engulfment of a photosynthetic cyanobacterium led to the development of the chloroplast (plastid).

Motility Symbiosis: Margulis also **proposed** that eukaryotic flagella and cilia originated from symbiotic associations with “spirochaete-like bacteria”



Prokaryotes or Eukaryotes?

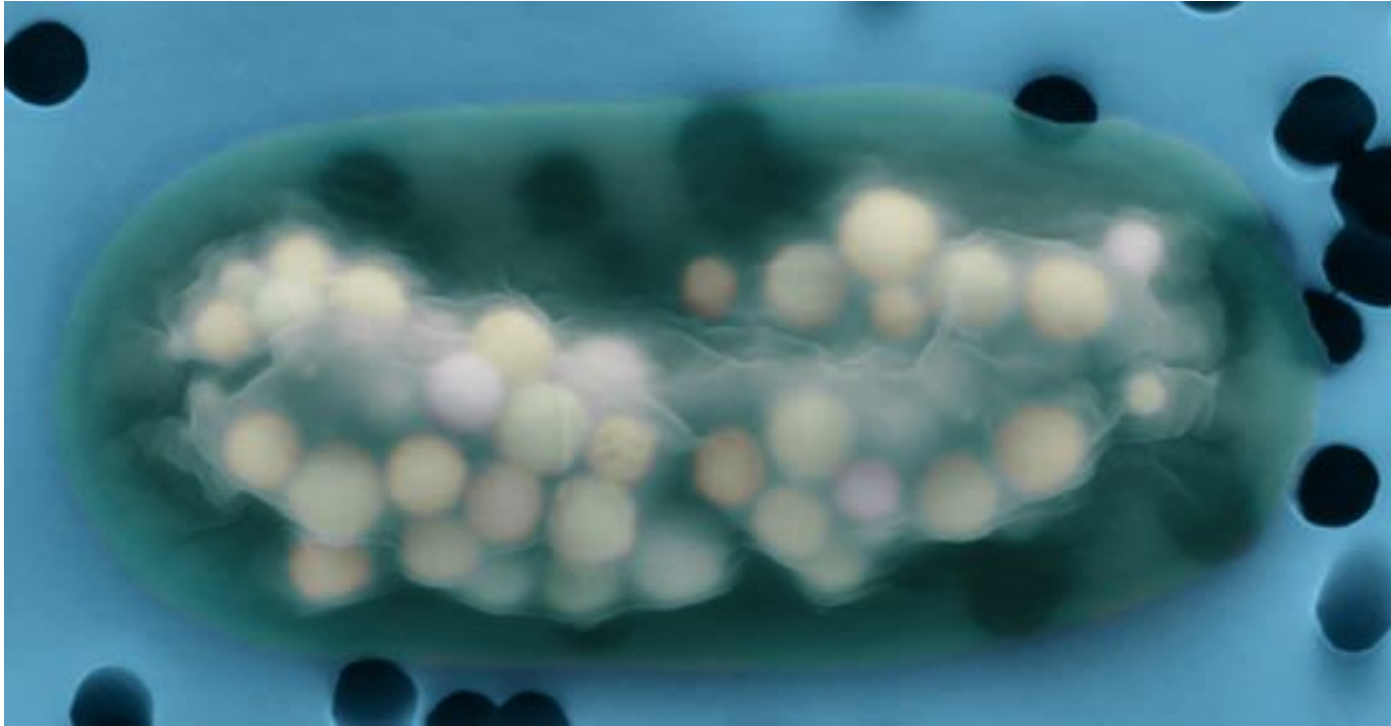


Prokaryotes or Eukaryotes?

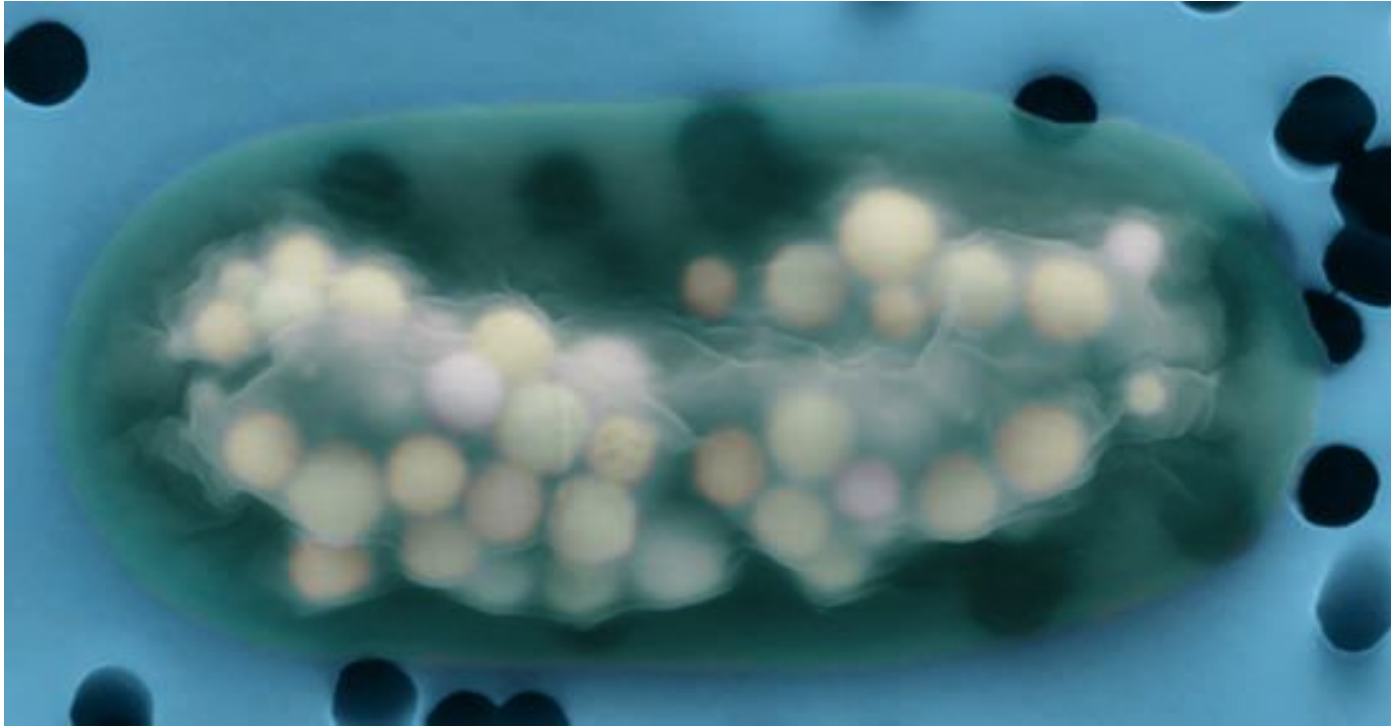


Bacterium *Rhodospirillum rubrum* with capsules

Prokaryotes or Eukaryotes?

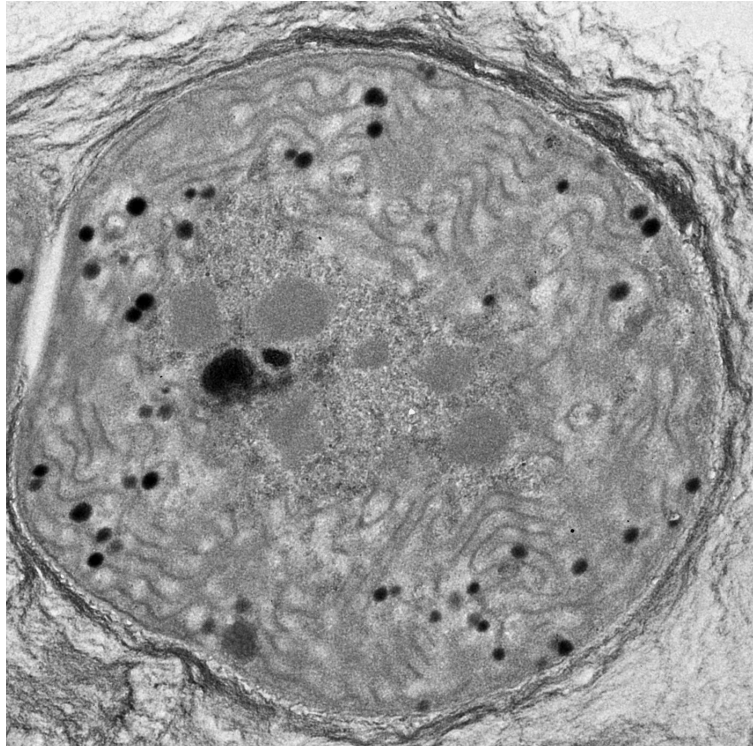


Prokaryotes or Eukaryotes?

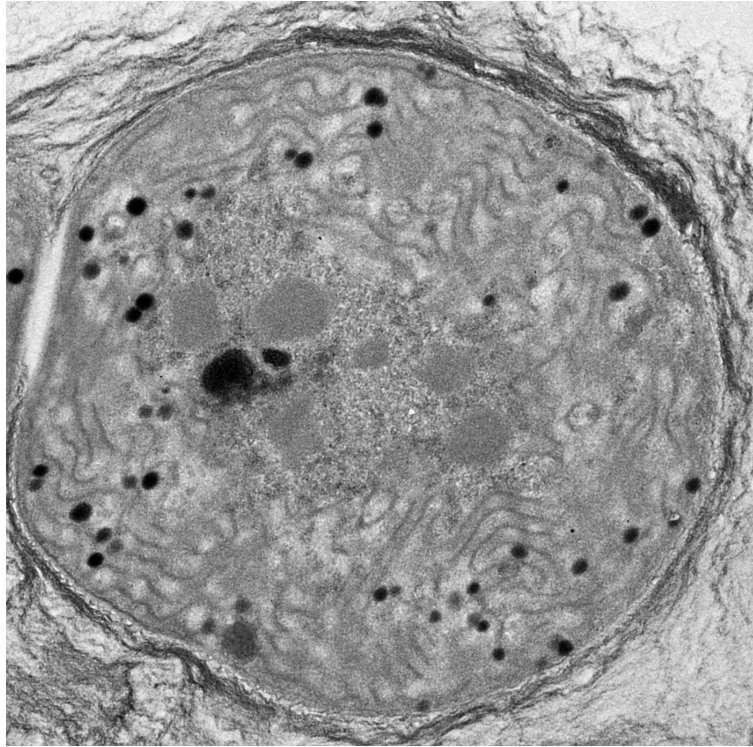


Cyanobacterium gloeomargarita containing granules of the carbonate mineral benstonite

Prokaryotes or Eukaryotes?

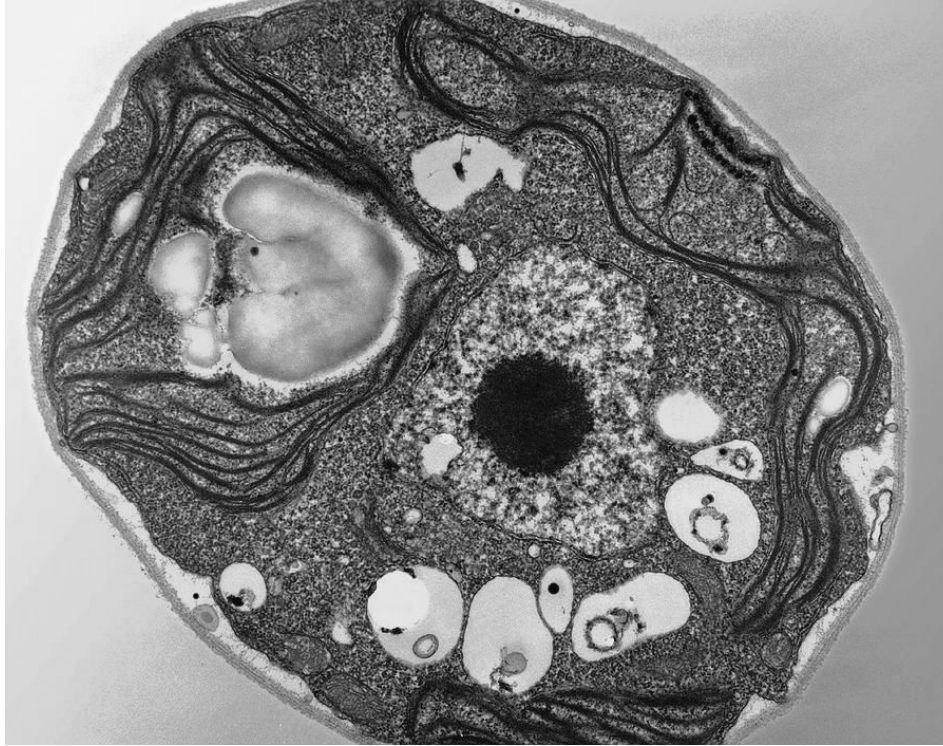


Prokaryotes or Eukaryotes?

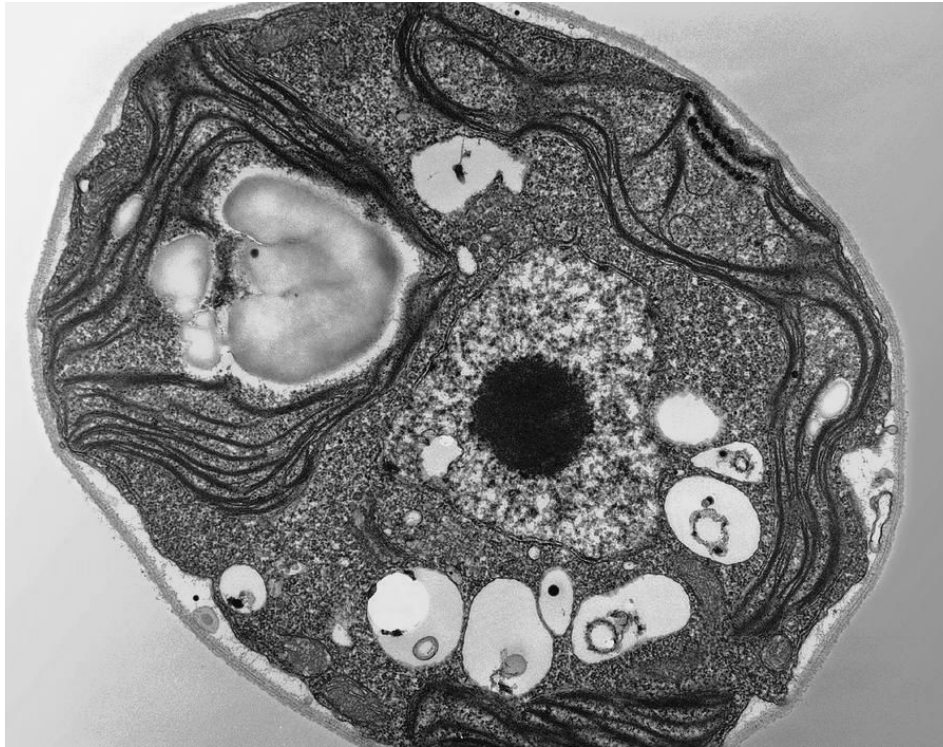


Filamentous nitrogen-fixing cyanobacterium *Anabaena*

Prokaryotes or Eukaryotes?

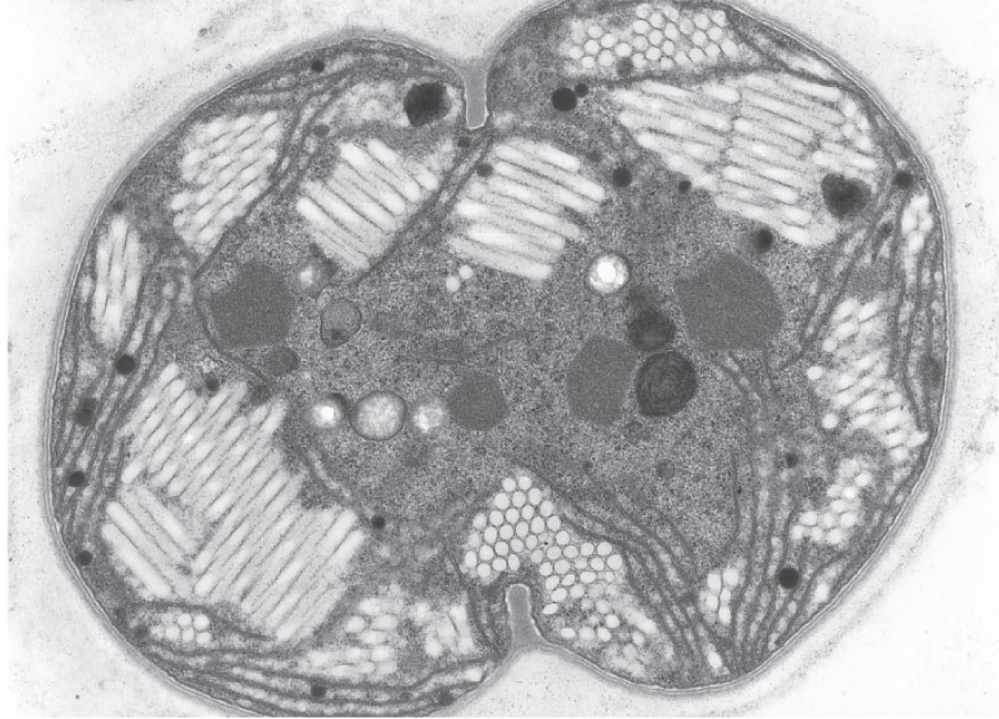


Prokaryotes or Eukaryotes?

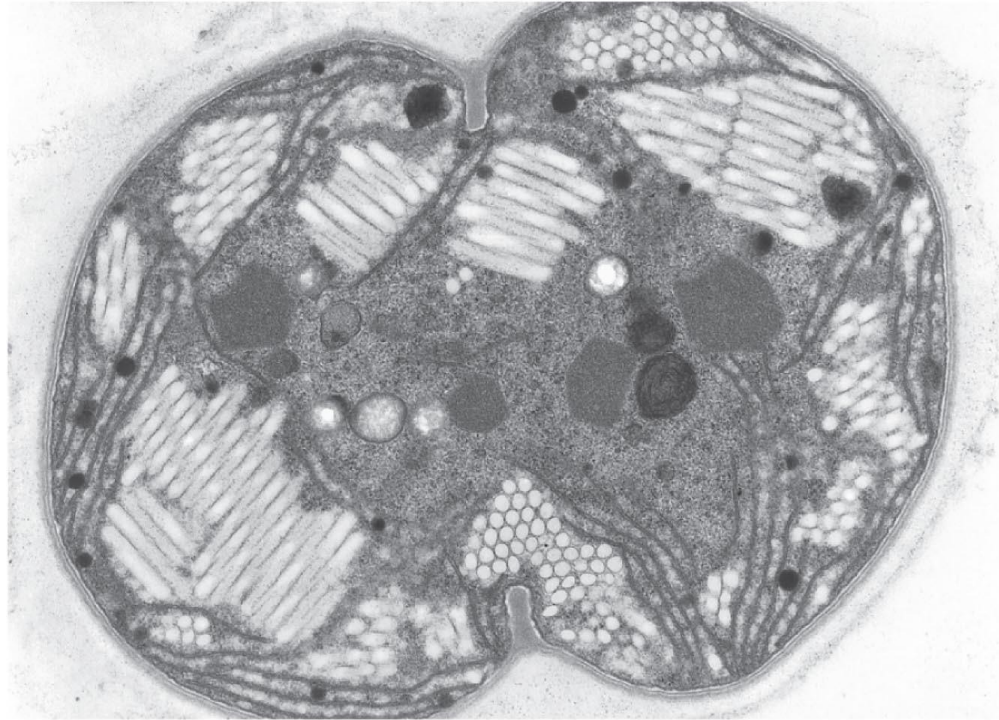


Single-cell green alga *Chlamydomonas reinhardtii*

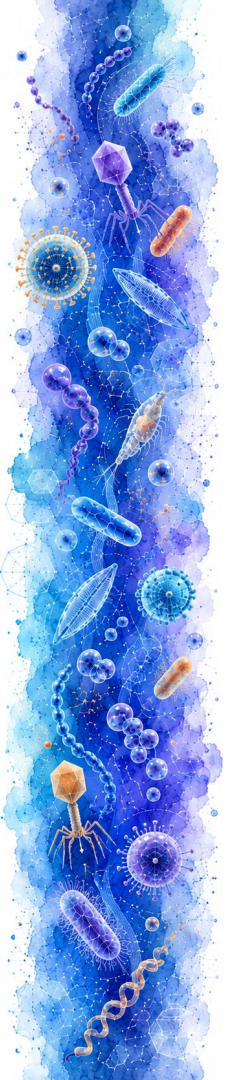
Prokaryotes or Eukaryotes?



Prokaryotes or Eukaryotes?



Cyanobacterium *microcystis* containing gas vesicles are arranged in bundles



This Week Read

Schleper, C., Rodrigues-Oliveira, T. Asgard archaea: have we found our microbial ancestors?.
EMBO J 45, 1836–1851 (2026). <https://doi.org/10.1038/s44318-026-00719-x>